Kristina P. Sinaga, PhD

Independent Researcher • PhD in Applied Mathematics
Machine Learning • Federated Learning • Data Science

kristinasinaga41@gmail.com | kristina01-hash.github.io | github.com/kristina01-hash | Google Scholar

RESEARCH PROFILE

Applied mathematician with a PhD from Chung Yuan Christian University (Taiwan) and postdoctoral experience at Chung Yuan Christian University and ISTI-CNR (Italy). Over seven years of experience in multi-view clustering, federated learning and privacy-preserving machine learning for heterogeneous data. Author of publications in IEEE, Elsevier and MDPI journals, with more than 3,870 citations (h-index 8). Seeking research positions in which mathematical methods for multi-source data are developed together with scientific applications.

RESEARCH INTERESTS

- **Clustering methodology:** k-means and fuzzy c-means algorithms for single-view and multi-view data, with learned view and feature weights and a focus on efficiency and accuracy for heterogeneous, high-dimensional data.
- **Pattern recognition and dimensionality reduction:** clustering-based feature selection and dimensionality reduction; scalable pattern recognition frameworks for high-dimensional data.
- **Federated learning and privacy-preserving AI:** adaptation of classical learning algorithms to multi-client, multi-view settings, with attention to privacy, data security and communication efficiency.

EDUCATION

PhD in Applied Mathematics
2016 – 2020

Chung Yuan Christian University, Taiwan

- **Dissertation:** Multi-View Fuzzy Clustering Algorithms for Multi-View Data (多視圖數據模糊聚類演算法)
- Graduated with high honors. Published 2 journal articles (J1, J2) and 1 conference paper (C1) during doctoral studies and presented at 1 international conference. Further journal articles arising from the doctoral research were published after graduation (J3, J4, 2021).

MSc in Mathematics
2013 – 2015

University of Sumatera Utara, Indonesia

- **Thesis:** Stochastic Optimization Models for Emergency Service Location Problems (*Model Optimasi Stokastik Penentuan Lokasi dan Jumlah Ambulan dengan Korelasi*).
- Research project on operations research, optimization theory, and mathematical modeling.

BSc in Mathematics
2008 – 2013

University of Sumatera Utara, Indonesia

- **Thesis:** Analysis of the Influence of Gross Regional Domestic Product, Education, and Unemployment on Poverty in Regencies and Cities in North Sumatra (*Analisis Pengaruh Produk Domestik Regional Bruto, Pendidikan dan Pengangguran terhadap Kemiskinan di Kabupaten/Kota Propinsi Sumatera Utara*)
- Graduated with honors. Research project on regional economic analysis using mathematical modeling.

RESEARCH AND ACADEMIC APPOINTMENTS

Independent Researcher
2020 – Present

Self-directed research, conducted between and alongside the appointments below

- Developed new approaches to multi-view clustering and federated learning, and prepared multiple manuscripts for submission to journals.
- Obtained an arXiv endorsement and published preprints.
- Authored one journal article (J8, *Electronics*), open-source software, and eight preprints (P1–P8) on clustering, federated learning, and other machine learning topics. In addition to my professional work, I have also contributed to various online publications through my personal blog, covering topics related to machine learning and other relevant subjects.

Postdoctoral Researcher October 2024 – September 2025
Institute of Information Science and Technologies (ISTI-CNR), Italy

- Designed and implemented new clustering algorithms for heterogeneous data, achieving 20–30% higher clustering accuracy on benchmark datasets.
- Authored five arXiv preprints (P2–P7) on clustering and other machine learning topics. Published a journal article (J7, *Electronics*) arising from the postdoctoral research in Taiwan, with ISTI-CNR and CYCU affiliations.
- Released two open-source Python packages, *mvkm-ed* and *gcomvkm* (see Open-Source Software), together with an archived code release (CS2) and a synthetic multi-view dataset (CS1).

Postdoctoral Fellow March 2023 – March 2024
Department of Applied Mathematics, Chung Yuan Christian University, Taiwan

- Developed single-view and multi-view clustering algorithms in federated and non-federated settings, including multi-view k-means (MVKM) and multi-view fuzzy c-means (MVFCM); one of these works led to a publication in *IEEE TPAMI*.
- Advanced the theoretical foundations of privacy-preserving machine learning and federated clustering.
- Provided MATLAB implementations for multi-source and multi-client data, and validated the methods experimentally on public multi-view datasets.

Lecturer Specialist S3 November 2020 – March 2022
Bina Nusantara University, Indonesia

- Taught mathematics and data analysis courses to more than 180 undergraduate and graduate students, with a reported 35% increase in student engagement.
- Supervised more than 10 student research projects; developed and updated curriculum materials; introduced project-based learning and real-world case studies.

TEACHING AND ACADEMIC SERVICE

Master in Information Systems Management 2020 – 2022
Bina Nusantara University

- Business Intelligence and Analytics (regular and online programs): data-driven decision making, statistical analysis and business intelligence tools; 100+ graduate students.

Bachelor in Computer Science 2021 – 2022
Bina Nusantara University

- Calculus I (2021) and Discrete Mathematics (2021–2022); 80+ undergraduate students.

Moderator, International Guest Lecturer Series 2020 – 2021
Moderated 4 international academic events

- Coordinated guest presentations and facilitated discussions among students and faculty from universities in Indonesia and abroad.

Outreach and Mentoring 2020 – 2022
Various institutions

- STEM outreach for underserved communities and educational institutions; science communication events and public lectures on mathematical concepts.
- Mentoring of students across institutions; support for diversity and inclusion initiatives.

PUBLICATIONS

8 journal articles (J1–J8), 4 conference papers (C1–C4), 8 preprints (P1–P8), 1 software release and 1 dataset (CS1–CS2), and the doctoral, master's and bachelor's theses. Total citations: 3,870+; h-index: 8. The labels used in this CV refer to the enclosed publication list, which gives full references, DOIs and code links and is also available on Google Scholar.

OPEN-SOURCE SOFTWARE

mvkm-ed 1.1.0 PyPI package
Multi-view clustering framework

- Combines Federated Multi-View K-Means (Fed-MVKM) (J6, *IEEE Transactions on Pattern Analysis and Machine Intelligence*) with Rectified Gaussian Kernel Multi-View K-Means (MVKM-ED) (P1, *arXiv*), a privacy-preserving distributed framework with enhanced discriminative power. Tested on synthetic data and the DHA (Depth-included Human Action) dataset.
- PyPI | GitHub

gcomvkm 0.1.0 PyPI package

Globally collaborative multi-view clustering

- Python implementation of Globally Collaborative Multi-View k-Means, which integrates collaborative transfer learning with entropy-regularized feature and view reduction (J7, *Electronics*). It dynamically eliminates uninformative components while balancing local view importance and global consensus. Tested on synthetic data.
- PyPI | GitHub

PEER REVIEW SERVICE

Reviewer since 2020 (having written +130 reviews, my record of which is visible on my ORCID: 0009-0000-6184-829X, but unfortunately I have no access to it and it cannot be repaired) for the following journals:

- **IEEE:** *IEEE Access*, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Knowledge and Data Engineering*.
- **Springer:** *Data Mining and Knowledge Discovery*, *Statistics and Computing*, *Journal of King Saud University – Computer and Information Sciences*, *The Journal of Supercomputing*.
- **ACM:** *ACM Transactions on Knowledge Discovery from Data*.
- **Elsevier:** *Computer Methods and Programs in Biomedicine*, *Information Fusion*, *Pattern Recognition*, *Neural Networks*, *Applied Soft Computing*, *Image and Vision Computing*, *Digital Applications in Archaeology and Cultural Heritage*.
- **Universal Wiser Publisher (UWP):** *Contemporary Mathematics*.
- **Intelligent Journal System (IJS):** *CMC: Computers, Materials & Continua*.
- **MDPI:** *Algorithms*, *Applied Sciences*, *Big Data and Cognitive Computing*, *Biomimetics*, *Designs*, *Diagnostics*, *Drones*, *Electronics*, *Entropy*, *Future Internet*, *Information*, *IoT*, *Journal of Imaging*, *Journal of Marine Science and Engineering*, *Machine Learning and Knowledge Extraction*, *Mathematics*, *Project Management Horizons*, *Sensors*, *Symmetry*.
- **Recognition:** MDPI *Mathematics* Exceptional Reviewers List, 2025.

AWARDS AND HONORS

- **2025** MDPI *Mathematics* Exceptional Reviewers List.
- **2020–present** Paper listed among the most-read documents in *IEEE Access*.
- **2020** Honorary Member, Phi Tau Phi Scholastic Honor Society, Chung Yuan Christian University, Taiwan.
- **2018** Japan Science and Technology Agency (JST) Scholarship, Niigata University, Japan.
- **2017** Japan Student Services Organization (JASSO) Scholarship, Niigata University, Japan.
- **2016** International Student Scholarship (full tuition and living expenses), Chung Yuan Christian University, Taiwan.

CERTIFICATIONS

Completed during a period of self-directed professional development (April 2022 – February 2023).

- **October 2022** Machine Learning Specialization, Coursera (Stanford University and DeepLearning.AI). Certificate
- **November 2022** TensorFlow Developer Specialization, Coursera (DeepLearning.AI). Certificate

TECHNICAL SKILLS

- **Programming and tools:** Python, R, MATLAB, Julia, C++, SQL, LaTeX, Git/GitHub, Docker, Linux/Unix.
- **Machine learning and scientific computing:** TensorFlow, PyTorch, scikit-learn, Keras, NumPy, Pandas, SciPy, Jupyter, Matplotlib, Seaborn.
- **Cloud and data infrastructure:** AWS, Google Cloud, Azure, Kubernetes, Apache Spark, Hadoop, MongoDB, PostgreSQL.

LANGUAGES

Bahasa Indonesia (native) • English (professional) • Mandarin (basic)

Last updated: October 2026

Publication List

Kristina P. Sinaga, PhD

Google Scholar | ORCID: 0009-0005-2230-8510 | GitHub

8 journal articles • 4 conference papers • 8 preprints • 1 software release and 1 dataset • doctoral, master's and bachelor's theses • 3,870+ citations • 12 code repositories. Citation counts are approximate (Google Scholar). Own name in bold. Last updated: October 2026.

JOURNAL ARTICLES

J8 **Multi-View Clustering Goes Federated: A Survey**
K. P. Sinaga
Electronics (MDPI), 15(18):4103, 2026.
DOI

J7 **A Globally Collaborative Multi-View k-Means Clustering**
K. P. Sinaga and M. S. Yang
Electronics (MDPI), 14(11):2129, 2025.
DOI | Code | Citations: 10+

J6 **Federated Multi-View K-Means Clustering**
M. S. Yang and **K. P. Sinaga**
IEEE Transactions on Pattern Analysis and Machine Intelligence, 47(4):2446–2459, 2025.
DOI | Code | Citations: 50+

J5 **Unsupervised Multiview Fuzzy c-Means Clustering Algorithm**
I. Hussain, **K. P. Sinaga**, and M. S. Yang
Electronics (MDPI), 12(21):4467, 2023.
DOI | Code | Citations: 40+

J4 **Collaborative Feature-Weighted Multi-View Fuzzy c-Means Clustering**
M. S. Yang and **K. P. Sinaga**
Pattern Recognition, 119:108064, 2021.
DOI | Code | Citations: 95+

J3 **Entropy K-Means Clustering with Feature Reduction under Unknown Number of Clusters**
K. P. Sinaga, I. Hussain, and M. S. Yang
IEEE Access, 9:67736–67751, 2021.
DOI | Code | Citations: 60+

J2 **Unsupervised K-Means Clustering Algorithm**
K. P. Sinaga and M. S. Yang
IEEE Access, 8:80716–80727, 2020.
DOI | Code | Citations: 3,380+

J1 **A Feature-Reduction Multi-View k-Means Clustering Algorithm**
M. S. Yang and **K. P. Sinaga**
IEEE Access, 7:114472–114486, 2019.
DOI | Code | Citations: 160+

CONFERENCE PAPERS

C4 **Analytics-Based on Classification and Clustering Methods for Local Community Empowerment in Indonesia**
D. Yuniati and **K. P. Sinaga**
International Conference on Soft Computing in Data Science (ICSCDS 2021), pp. 133–145, 2021.
DOI

C3 **Telecommunication Analytics Based on Customer Segmentation Using Unsupervised Algorithms**
H. Wibowo and **K. P. Sinaga**
2021 3rd International Conference on Cybernetics and Intelligent System (ICORIS 2021), pp. 1–6, 2021.
DOI | Citations: 3

C2 **Machine Learning Approaches for Marketing Campaign in Portuguese Banks**
J. Alexandra and **K. P. Sinaga**
2021 3rd International Conference on Cybernetics and Intelligent System (ICORIS 2021), pp. 1–6, 2021.
DOI | Citations: 8

C1 **Modified Relational Mountain Clustering Method**
K. P. Sinaga, J. N. Hsieh, J. B. M. Benjamin, and M. S. Yang
Artificial Intelligence and Soft Computing: 17th International Conference (ICAISC 2018), pp. 690–701, 2018.
DOI | Citations: 3

PREPRINTS

P8 **Multi-View Clustering Goes Federated: A Survey**
K. P. Sinaga
Qeios preprint of J8.
DOI

P7 **Calibration Meets Reality: Making Machine Learning Predictions Trustworthy**
K. P. Sinaga and A. S. Nair
arXiv:2509.23665, 2025.
DOI | Code | Citations: 4

P6 **Disproving the Feasibility of Learned Confidence Calibration Under Binary Supervision: An Information-Theoretic Impossibility**
A. S. Nair and **K. P. Sinaga**
arXiv:2509.14386, 2025.
DOI

P5 **Personalized Federated Heat-Kernel Enhanced Multi-View Clustering via Advanced Tensor Decomposition Techniques**
K. P. Sinaga
arXiv:2509.16101 (v4), 2025.
DOI | Citations: 1

P4 **FedHK-MVFC: Federated Heat Kernel Multi-View Clustering**
K. P. Sinaga
arXiv:2509.15844 (v2), 2025.
DOI | Code | Citations: 5

P3 **Parameter-Free Entropy-Regularized Multi-View Clustering with Hierarchical Feature Selection**
K. P. Sinaga, S. Colantonio, and M. S. Yang
arXiv:2508.05504, 2025.
DOI | Code | Citations: 1

P2 **Bi-Level Multi-View Fuzzy Clustering with Exponential Distance**
K. P. Sinaga and S. Colantonio
arXiv:2503.22932, 2025.
DOI | Code | Citations: 2

P1 **Rectified Gaussian Kernel Multi-View k-Means Clustering**
K. P. Sinaga
arXiv:2405.05619, 2024.
DOI | Code | Citations: 5

SOFTWARE AND DATASETS

CS2 **KristinaP09/calibration-analysis: CA: Binary Classification (Version v1.0.0)**
K. P. Sinaga
Computer software, Zenodo, 2025.
DOI

CS1 **Synthetic Multi-View Cardiovascular Dataset for Federated Clustering**
K. P. Sinaga
Dataset, IEEE DataPort, 2025.
DOI

DISSERTATION AND THESES

D1 **Multi-View Fuzzy Clustering Algorithms for Multi-View Data (多視圖數據模糊聚類演算法)**
K. P. Sinaga (許蒂娜)
PhD dissertation, Department of Applied Mathematics, Chung Yuan Christian University, Taiwan, pp. 1–92, 2020.
DOI

M1 **Stochastic Optimization Models for Emergency Service Location Problems**
K. P. Sinaga
MSc thesis (original title: Model Optimasi Stokastik Penentuan Lokasi dan Jumlah Ambulan dengan Korelasi), Department of Mathematics, University of Sumatera Utara, Indonesia, 2015.

B1 **Analysis of the Influence of Gross Regional Domestic Product, Education, and Unemployment on Poverty in Regencies and Cities in North Sumatra**
K. P. Sinaga
BSc thesis (original title: Analisis Pengaruh Produk Domestik Regional Bruto, Pendidikan dan Pengangguran terhadap Kemiskinan di Kabupaten/Kota Propinsi Sumatera Utara), Department of Mathematics, University of Sumatera Utara, Indonesia, pp. 1–68, 2013.